

Pitfalls and Hurdles II

Psychological Hurdles

- ☐ Mental Attitude
 - ☐ Research Impotence
 - ☐ Dealing with Criticism
 - ☐ Early Morning -- Cold Start
 - ☐ Theorem Envy
 - ☐ Fear of Exposure
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Mental Attitude

- Part of a researcher's skill includes an appropriate mental attitude to his/her work.
 - This can be learnt, if you know what you are aiming for and are determined enough.
 - One of the main ingredients of this mental attitude is a belief in what you are doing.
 - You must believe that the problem you are tackling is important and that your contribution to the solution is significant. Otherwise, how are you to generate the energy to see yourself through the long hours of hard work required?
 - The first step in obtaining this self assurance is to pick a research topic you can believe in.
 - Of course, you must not become so arrogant that you can no longer listen to criticism.
 - You must be prepared to modify your ideas if they are wrong.
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Research Impotence

- ❑ Doubts about your own ability can put you in a frame of mind where the dedication and enthusiasm necessary to produce results evaporates.
 - ❑ Edison said that genius was 1% inspiration and 99% perspiration, and he should know!
 - ❑ The way out of this vicious circle is to realise that research ability does not depend on some magic essence. It is a skill which can be learnt, like any other.
 - ❑ You too can do original research
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Dealing with Criticism

- If you are to make progress in your research you will have to learn to seek out criticism and take it into account.
 - You will have to learn to differentiate between valid and invalid criticism.
 - If you feel too close to the subject to decide, ask a friend for a second opinion.
 - If the criticism is invalid, maybe the critic has misunderstood. Can you improve your explanation?
 - You are going to have to learn to take some knocks: rejections from journals; rough rides in question time. Take it with a smile. Learn what you can. Don't be tempted to give up -- you are in good company.
 - If you study the lives of famous scientists you will see that many of them had to endure very heavy criticism.
 - In fact some of the best work is the product of personal feuds – each scientist busting to outdo the other.
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Early Morning -- Cold Start

- Make yourself a regular working schedule -- and stick to it. It doesn't have to be 9--5, but there should be a definite time of day when you expect to start work.
 - Otherwise, you will find yourself postponing the evil moment with endless, routine, domestic chores.
 - Make sure you leave some non-threatening, attractive task to do first thing.
 - For instance, do not leave off writing the day before at the beginning of a new hard section.
 - Leave something easy to start writing: a paragraph which is routine for you or a diagram to draw.
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Theorem Envy

- You have chosen a new field where the research methodology has not yet been worked out. You may get a hankering for the methodology you were brought up on.
 - For mathematicians this might be the longing to prove clean, clear theorems -- theorem envy.
 - Be wary! Only try to prove a theorem if it is clearly relevant to your overall purpose.
 - For instance, proving the termination of a procedure you have found to be useful may well be relevant.
 - Finding a procedure whose termination you can prove, but which is not otherwise interesting, is not relevant.
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Fear of Exposure

- ❑ You have a great idea and it only remains to test it by proving a theorem, writing a program, explaining it to a friend etc. But something is holding you back. You find it difficult to start work.
 - ❑ Could it be that you are secretly afraid that your idea is not so great after all?
 - ❑ Hard experience has taught you that ideas that appear to be solutions to all your problems in the middle of the night evaporate in the cold dawn.
 - ❑ Courage! Research is always like this. Ten steps forward and nine steps back.
 - ❑ The sooner you subject your idea to the acid test, the sooner you will discover its limitations and be ready for the next problem.
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